Closed Relative Orbit Design in Dawn–Dusk SSO

16.346 - Astrodynamics

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1 Introduction - AP, AA

Dawn–dusk Sun-synchronous orbits (SSOs) provide near-continuous solar illumination, making them ideal for distributed orbital systems such as space-based data centers. This project investigates a three-satellite formation consisting of a chief spacecraft and two deputies in a dawn–dusk SSO. The objective is to design relative orbits that remain bounded over long durations while minimizing control effort.

The project, as described by the project handout from [1] assumes that a reference node satellite (hereafter referred to as the "chief") will orbit the Earth in proximity to two compute or storage modules (hereafter referred to as the "deputies"). This architecture allows for inter-satellite communication, power sharing, and predictable operational conditions.

The scope of this project will be the behavior of the satellites **after** they have been put into orbit. While we do explore station-keeping in terms of what forces are necessary to maintain the orbits, what equipment is used to perform these maneuvers is outside of the scope of this project. This also means that concept of operations of the orbit establishment is assumed handled.

This project will explore two different designs: Design A has a deputy-chief separation range of 0.5 - 1.5 km, while design B stretches that to a range of 2 - 5 km. Additionally, the bulk of this analysis will be done in the Hill frame, so we shall discuss velocity and position in the radial, in-track, and cross-track directions of each satellite.

The parameters we look to optimize are the distances between the chief and deputy over time, and the relative stability of their respective orbits despite external forces.

The remainder of the project follows this outline, covering all three phases of the project therein:

The Methodology discusses the approach we took and the tools we used.

The Analysis section discusses how the parameters specific to this case study were calculated

The Results & Discussion section shows the results found for the specific case studies and discusses the insights provided by the results

The Summary and Future Work section provides a summary of the findings, limitations, and the path for future work

2 Methodology

Implementation and Metrics

Our simulations are conducted using Orekit's python wrapper, propagating out mean elements in order to analyze the long-term effects of different perturbations and mitigating numerical error propagation.

For our simulations, we reuse the same initial conditions for the chief and propagate over the same six-month period, starting on January 1, 2026. The bulk of our analysis takes places in the Hill frame to visualize geometric distortion of each formation. Further, our metrics of interest are solar power harvested, chief-deputy separations over time, and the asymmetry of the the deputies' positions relative to the chief. We take special interest in the asymmetry, given its influence on station-keeping complexity and long-term formation feasibility.

2.1 Idealized Force Model - AP

Considerations and Strategy - AP

The idealized model assumes a circular chief orbit and neglects perturbations beyond two-body gravity i.e. J_2 Perturbations. This enables the use of linearized relative motion equations (HCW), which are valid for small separations relative to the orbital radius, to establish an initial state. The primary strategic goal is to design initial conditions that yield bounded relative motion while satisfying separation constraints. This model provides intuition and a baseline for comparison with higher-fidelity dynamics.

Approach - AP

2.1.1 Linearized Relative Motion (HCW Approximation)

For small separations, the relative motion between a deputy and the chief can be approximated using the Hill–Clohessy–Wiltshire (HCW) equations in the radial–in-track–cross-track (RTN) frame:

$$\ddot{x} - 2n\dot{y} - 3n^2 x = 0 \tag{1}$$

$$\ddot{y} + 2n\dot{x} = 0 \tag{2}$$

$$\ddot{z} + n^2 z = 0 \tag{3}$$

where $n = \sqrt{\mu/a^3}$ is the mean motion of the chief orbit.

These equations provide insight into the natural relative motion structure. Equations 1 & 2 show that in-track motion is coupled with radial motion. Equation 3 shows that cross-track motion is decoupled and harmonic. Lastly, bounded motion requires $\delta a = 0$.

Although the HCW equations neglect J_2 perturbations, they are useful for constructing initial guesses for relative orbit geometries, which are subsequently validated using the full J_2-perturbed dynamics.

The chief orbit is chosen to satisfy the Sun-synchronous condition, which arises from J_2-induced nodal precession. The rate of change of the right ascension of the ascending node is given by

$$\dot{\Omega} = -\frac{3}{2} J_2 \sqrt{\frac{\mu}{a^3}} \left(\frac{R_E}{a(1-e^2)} \right)^2 \cos i. \tag{4}$$

For a dawn–dusk SSO, $\dot{\Omega}$ is selected to match the apparent motion of the Sun (approximately one revolution per year). From that and our selection of an altitude of $1,000$ km, our choice of inclination is $\approx 99°$.

2.1.2 J_2-Invariant Formation Design

To ensure long-term stability of the formation under J_2 perturbations, the chief and deputies are required to share the same mean orbital elements:

$$a_c = a_d, \quad e_c = e_d, \quad i_c = i_d. \tag{5}$$

Under this assumption, the secular rates satisfy

$$\dot{\Omega}_c = \dot{\Omega}_d, \quad \dot{\omega}_c + \dot{M}_c = \dot{\omega}_d + \dot{M}_d, \tag{6}$$

ensuring that the relative geometry remains bounded over time. This mirrors the J_2-invariant condition used in the geosynchronous analysis but is more restrictive due to the stronger perturbations present in low Earth orbit.

2.1.3 Deputy Parameterization

The Sun-synchronous constraint fixes the chief's inclination and limits variation in orbital elements. Therefore, the deputy orbits are parameterized using relative orbital elements (ROEs).

The deputies are initialized such that

- $\delta a = 0$ (no relative drift in semi-major axis),
- $\delta e_x, \delta e_y$ define in-plane motion,
- $\delta i_x, \delta i_y$ define cross-track motion,
- δM defines along-track separation.

The initial position of each deputy is constructed as

$$\vec{r}_d(0) = \vec{r}_c(0) + \vec{d}, \tag{7}$$

where $\vec{d}$ is derived from the chosen set of relative orbital elements.

2.1.4 Planned Formation Geometries

Regardless of the variable distance constraints between designs A and B, we aim to maintain a bounded relative orbit initialized such that orbital perturbations have minimized effects. By doing so, we improve our Δv budgets when applying a control scheme to each design. Further, we aim for relative orbits which degrade symmetrically, ensuring even propellant loads on each deputy.

Along-track ("string of pearls") Deputies are only separated in the along-track direction; this configuration is highly robust under J_2. This is the more convenient choice for Design B as the significantly looser distance requirements permit the deputies to occupy elliptical orbits ahead and behind of the chief, as seen in Figure 1. Importantly,

Helix about Chief Design A requires tighter formation flying, which would be required in cases of inter-satellite servicing. [4] derive a method for formation flight by having two deputies form a helix around the chief, remaining on opposite ends of it by

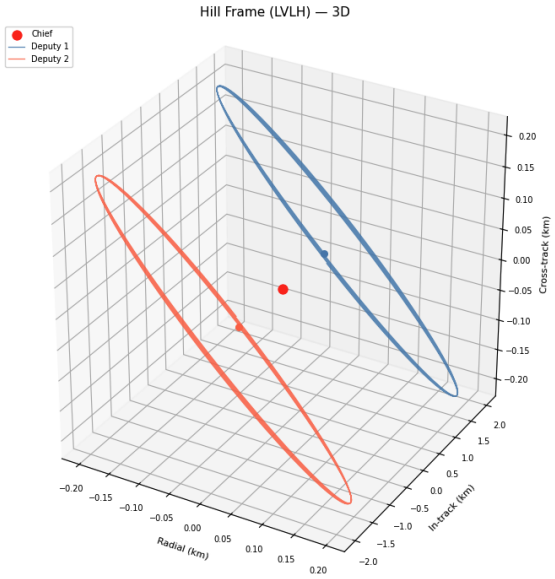


Figure 1: String of pearls setup for the chief-deputy formation. By loosening the distance requirements, we can set up relative orbits centered ahead and behind of the chief.

enforcing antiparallel $\vec{e}$, $\vec{i}$ vectors between the chief and deputies. Figure 2 demonstrates a short-term propagation of this orbit.

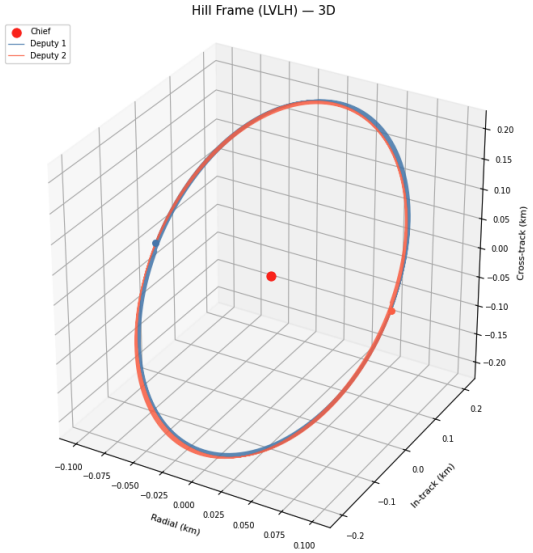


Figure 2: Helix setup for the chief-deputy formation. Servicing missions require means of remaining close to chief without risk of inter-satellite collisions.

When J_2 invariant initial conditions are selected, both of these mission designs have predictable, symmetric, and slow decay.

2.2 Full Force Model – AA, AP

Considerations and Strategy

The full force model was developed to evaluate the long-term stability of the satellite formations under a more realistic orbital environment than the idealized J_2-only case. While the idealized model provides useful intuition and a tractable baseline, it neglects several perturbations that become significant over multi-month propagation intervals in low Earth orbit. Building upon the initial relative orbit designs, the following additional perturbations are incorporated:

- Higher-order Earth gravity harmonics
- Solar radiation pressure
- Atmospheric drag
- Third-body effects from the Sun and Moon

We anticipate significantly greater secular drift after the addition of these forces. Beyond exposing the limitations of the idealized model, this drift directly motivates the control schemes recommended for station-keeping in the Impulsive Stationkeeping section. Impulsive maneuvers are considered first, but they are evaluated against low-thrust alternatives such as electric propulsion. In optimizing our control scheme, we also aim to quantify which perturbation contributes most to the required corrective effort.

2.3 Impulsive Stationkeeping

Considerations and Strategy

The chief is assumed to have its own stationkeeping already handled, so our maneuver schemes focus on imposing the deputies' relative orbital parameters to be within some error bound. That is, stationkeeping is done assuming the chief is a noncooperative agent.Thus, all additional burns must be planned and executed by the deputies.

Because the majority of secular drift in the full force model is caused by drag, it's necessary to perform stationkeeping burns while the effects of drag are minimal in order to extend formation lifetime. The helix formation will require a higher frequency of stationkeeping burns due to the lower tolerance of the helix formation to perturbations while the string of pearls will require infrequent burns primarily to resolve in-track drift.

We select our control schemes such that computational requirements are minimized, which allows for greater emphasis on payload computations during an actual mission.

Furthermore, in either scheme we aim to keep our cumulative Δv on the order of 1 m/s because of how weakly the additional perturbations affect our orbits in the short term.

Control Schemes

Design A

The helix formation first presented in [4] was successfully demonstrated by the AVANTI mission, as summarized in [2]. Motivated by the demonstration's success, we reuse their control scheme for the stage of the mission wherein AVANTI became a deputy to another satellite.

We first define dimensionless relative orbital elements according to

$$\begin{aligned} \delta\boldsymbol{\alpha} &= f(\boldsymbol{\alpha}, a_c, i_c) - f(\boldsymbol{\alpha}_c, a_c, i_c) \\ &= (\delta\alpha, \delta\lambda, \delta i_x, \delta i_y, \delta e_x, \delta e_y)^\intercal \end{aligned} \tag{8}$$

where

$$f(\boldsymbol{\alpha}, a_c, i_c) = (a/a_c, u + \Omega\cos i_c, i, \Omega\sin i_c, e\cos\omega, e\sin\omega) \tag{9}$$

and the subscript c denotes a parameter of the chief. With this we can continue by formulating the dimensional state estimate as

$$\boldsymbol{x} = (a\delta\dot{a}, a\delta\boldsymbol{\alpha})^\intercal \tag{10}$$

which is particularly useful because $\dot{a}$ encodes the effects of differential drag into the state, simplifying the necessary calculations. It should also be noted that this formulation allows us to compute the necessary state transitions analytically. To do so, we define the state at time t using the initial conditions and state transition matrix $\boldsymbol{\Phi}(t, t_0)$:

$$\begin{aligned} \boldsymbol{x} &= \boldsymbol{\Phi}(t, t_0)\boldsymbol{x}(t_0) \\ &= \begin{bmatrix} \mathbf{I}_{1\times1} & \mathbf{0}_{1\times6} \\ \tilde{\boldsymbol{\Phi}}_{d-\text{drag}}(t, t_0) & \boldsymbol{\Phi}_{\text{HCW}} + \boldsymbol{\Phi}_{J2}(t, t_0) \end{bmatrix} \boldsymbol{x}_0 \end{aligned} \tag{11}$$

wherein "HCW" denotes the Hill-Clohessy-Wiltshire contribution, "J2" encodes the mean effect of Earth's oblateness, and "d-drag" encodes the joint effects of differential drag. Because we found J2 and drag to be the dominant perturbing forces in our full-force analysis, we can neglect other perturbations in our control scheme. This will prove useful in simplifying the necessary computations and the stationkeeping remains effective, as can be seen in the results section for stationkeeping.

It should be noted that [2] uses the following control scheme plan:

$$\boldsymbol{x}_F = \boldsymbol{\Phi}(t_F, t_0)\boldsymbol{x}_0 + \boldsymbol{\Phi}(t_F, t_1)\Delta\boldsymbol{x}_1 + \cdots + \boldsymbol{\Phi}(t_F, t_m)\Delta\boldsymbol{x}_m \tag{12}$$

which is derived in [3] and functions by planning out multiple impulsive burns in advance so as to work around periods of inactive thrust (e.g., when docked with another satellite). The yet undefined maneuvers, $\Delta\boldsymbol{x}_m$, are accumulated throughout the simulation in order to reach a given desired state, $\Delta\boldsymbol{x}_F$. It should be noted that such dead periods of thrust are excluded from the scope of this report, and therefore our estimates underestimate the actual Δv cost of the station keeping, especially when coupled with the servicing maneuvers left unexplored in this report.

Regardless, the cumulative Δv cost for performing a sequence m maneuvers can be calculated by the following quadratic relation:

$$J_{\text{plan}} = \sum_{i=1}^{m} \left(\|a\Delta\delta\boldsymbol{i}\|\right)_i^2 + \sum_{i=1}^{m} \left(\|a\Delta\delta\boldsymbol{e}\|\right)_i^2 + \sum_{i=1}^{m} \left(\|a\Delta\delta a\|\right)_i^2 + \sum_{i=1}^{m} \left(\|a\Delta\delta\lambda\|\right)_i^2 \tag{13}$$

where $\boldsymbol{e}, \boldsymbol{i}$ are the relative eccentricity and inclination vectors. This formulation for J_{plan} is convenient because it has a closed form, analytical solution. For brevity, it can be found in appendix A of [2]. In practice, because the antiparallel relative eccentricity/inclination phasing characteristics are the target for each deputy and their burns take place after small deviations from the nominal state, the transfers don't risk inter-satellite collisions.

Design B

Because the string of pearls is initialized with $\delta a = 0$, differential drag takes significantly longer to take hold and therefore, smaller and less frequent burns are permitted for stationkeeping. Furthermore, because the string of pearls conserves e, i, a, ω and Ω between the chief and deputies, burns can be used exclusively for stationkeeping [2] as pro/retrograde burns raise and lower the energy of orbit, combating δa drift alongside $\delta\lambda$ drift. Though differential drag can take hold for differing geometries, it's already assumed that the chief and deputies are geometrically identical.

Drag results in drift in the in-track, T, direction at a rate given by

$$\dot{T} = a\delta\dot{\lambda} = -\frac{3}{2}n\delta a \tag{14}$$

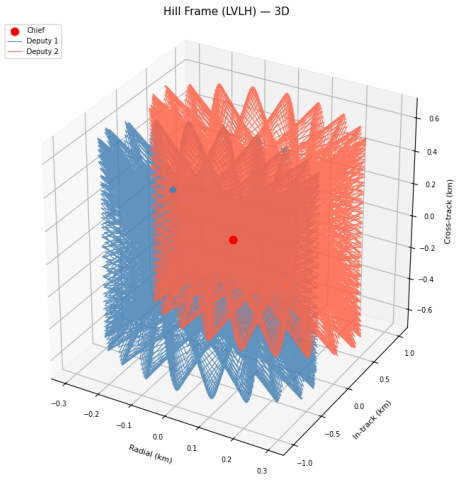
where n is the mean motion of the chief. It should be noted that controlling for the in-track drift, which also controls the semi major axis, has the practical application of allowing for a streamlined sensor suite [2].

Beyond this, no additional control is required and each deputy can read its current in-track position, and if it's at or passed the pre-defined allowed distance between the chief and deputies, 5 km, a burn is executed. In practice, this will result in a "saw-tooth" shape for the chief deputy separations as the deputies drift between burns.

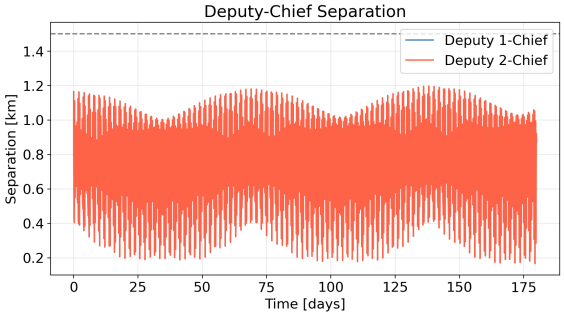
3 Results

Idealized Force Model - AP

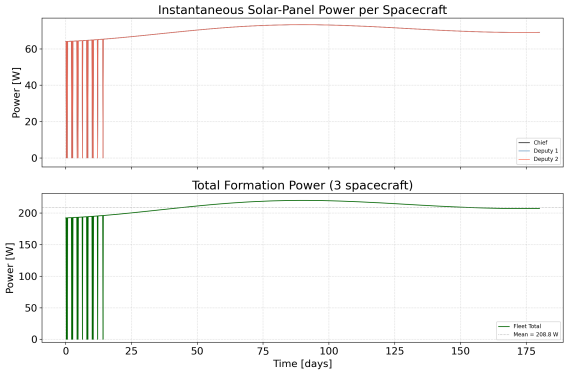
Design A



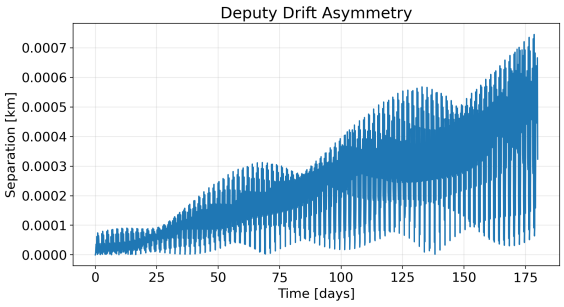
(a) Hill-frame trajectory of the Design A helix formation under J_2 perturbations over six months.



(b) Chief–deputy separation distance for Design A over six months.



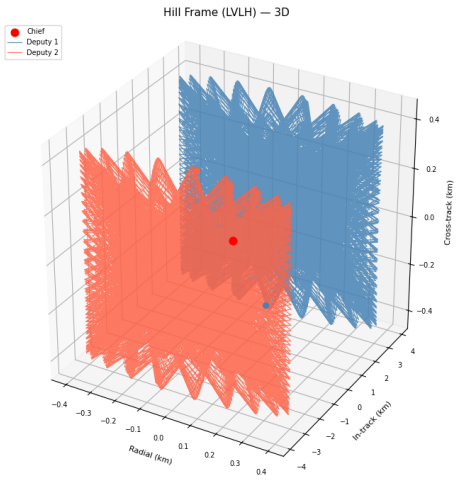
(c) Solar power input history for the Design A configuration.



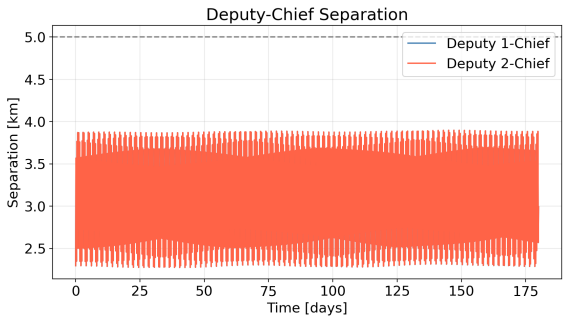
(d) Asymmetry growth between the two deputies in Design A.

Figure 3: Idealized force model results for Design A under J_2 perturbations over six months.

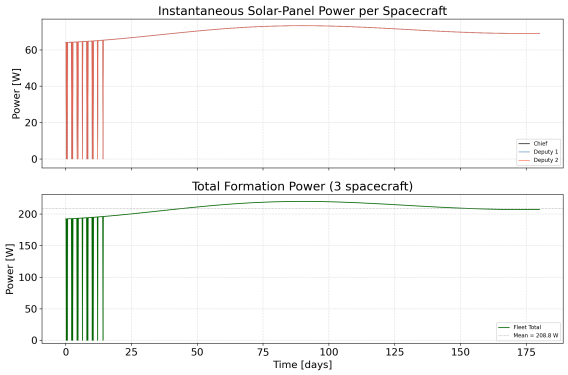
Design B



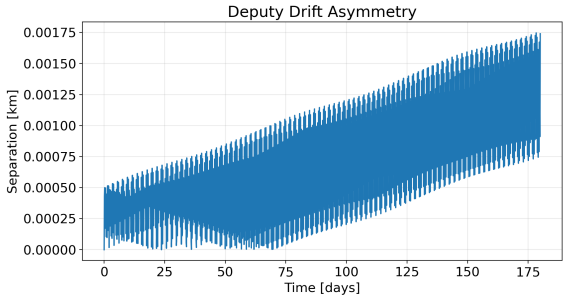
(a) Hill-frame trajectory of the Design B string-of-pearls configuration.



(b) Chief–deputy separation distance for Design B.



(c) Solar power input history for the Design B configuration.



(d) Deputy asymmetry evolution for Design B.

Figure 4: Idealized force model results for Design B under J_2 perturbations over six months.

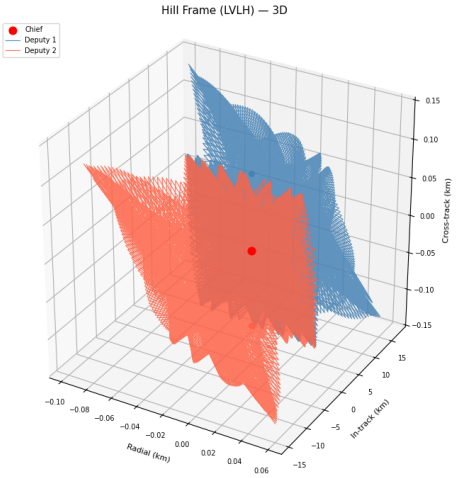
The idealized simulations were used to evaluate the long-term behavior of the two proposed formation geometries under two-body gravity with J_2 perturbations. Figure 4 show the six-month propagation of the string-of-pearls configuration corresponding to Design B. The deputies remain separated primarily in the along-track direction and follow bounded, quasi-periodic relative trajectories about the chief. Although gradual secular drift is observed over time, the two deputies exhibit nearly identical separation histories and relative motion geometry, indicating that the selected relative orbital element parameterization successfully minimized differential perturbation effects between spacecraft. Because the allowable separation distances for Design B are relatively large, the configuration remains operationally robust despite the slow drift induced by J_2 per-

turbations. In contrast, Figure 3 shows that the helix configuration used for Design A is significantly more sensitive to perturbations due to its tighter formation constraints. The deputies initially maintain the desired three-dimensional helical motion about the chief, but even small secular deviations eventually become significant relative to the smaller allowable bounding region.

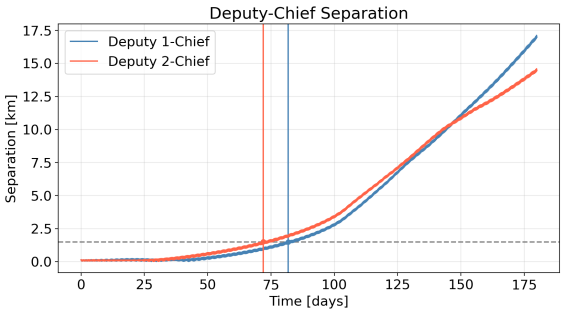
The Hill-frame projections further demonstrate that both formations preserve highly structured relative motion throughout the propagation interval. In the radial–in-track plane, the deputy trajectories form closed elliptical patterns characteristic of bounded HCW motion, while the in-track–cross-track projections exhibit the expected oscillatory behavior associated with the imposed relative eccentricity and inclination vectors. Figure **??** also shows the evolution of the chief spacecraft orbital elements under J_2 perturbations over six months. As expected for a Sun-synchronous orbit, the right ascension of the ascending node undergoes steady secular precession while the remaining orbital elements remain nearly constant due to the use of mean elements in the propagation model. Overall, the idealized simulations demonstrated that both formation geometries could maintain bounded and symmetric relative motion under J_2 perturbations alone, with the string-of-pearls configuration providing greater tolerance to secular drift and the helix configuration offering improved close-proximity behavior at the cost of increased sensitivity to perturbations.

Full Force Model - AP, AA

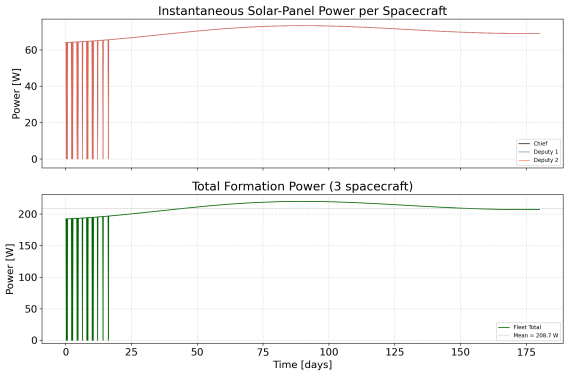
Design A



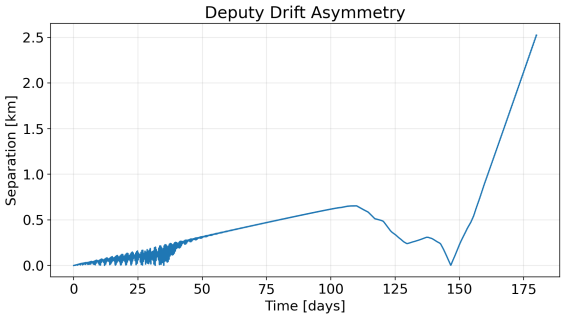
(a) Hill-frame trajectory of Design A under the full force perturbation model.



(b) Chief–deputy separation distance for Design A under the full force model.



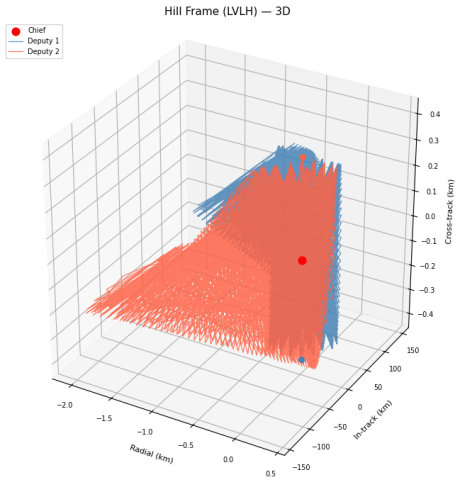
(c) Solar power input history for Design A under the full force model.



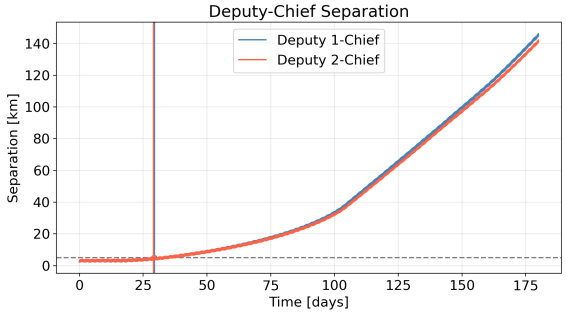
(d) Deputy asymmetry evolution for Design A under the full force model.

Figure 5: Full force model results for Design A over six months.

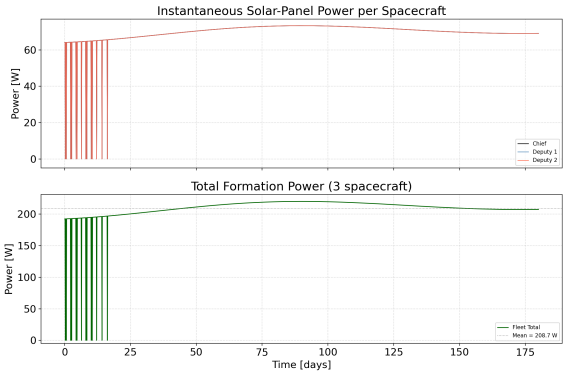
Design B



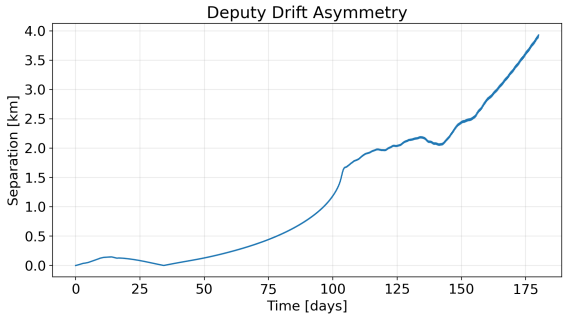
(a) Hill-frame trajectory of Design B under the full force perturbation model.



(b) Chief–deputy separation distance for Design B under the full force model.



(c) Solar power input history for Design B under the full force model.



(d) Deputy asymmetry evolution for Design B under the full force model.

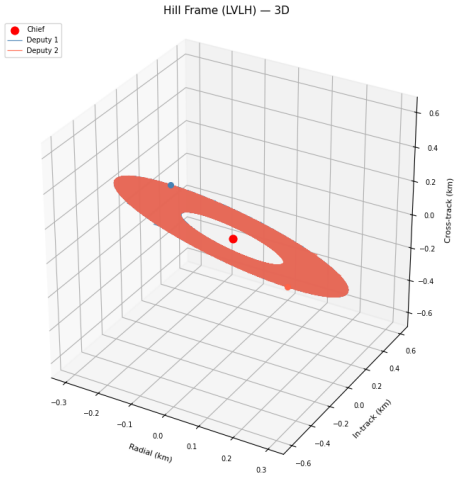
Figure 6: Full force model results for Design B over six months.

Incrementally adding the additional perturbations discussed in the methodology section almost unilaterally yielded negligible changes in the relative trajectories of the deputies. However, the addition of drag led to quadratically growing secular drift, per $D = \frac{1}{2}\rho v^2 C_d A$.

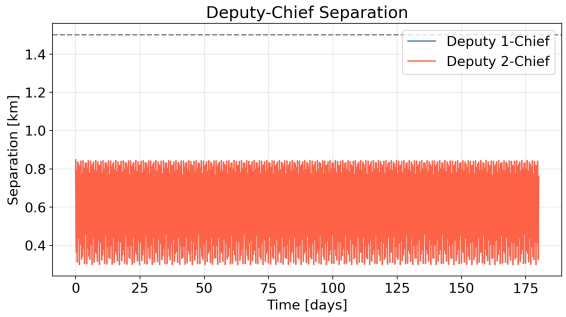
Visually, we observe the non-conservative effects of drag forcing the deputy trajectories to resemble "wings" about the chief, per figures 5a and 6a. Additionally, we notice that the helix formation is significantly more impacted as the imposed differences in $\delta\boldsymbol{e}, \delta\boldsymbol{i}$ facilitate differential drag.

Impulsive Stationkeeping - AA

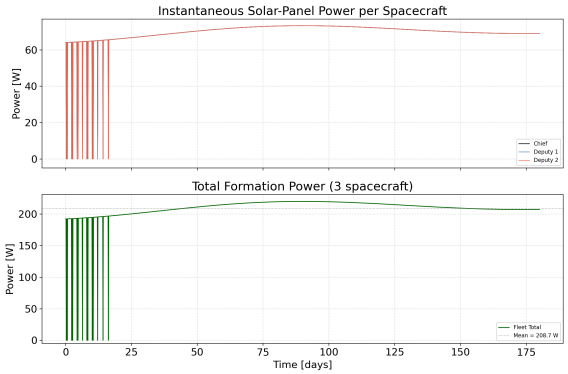
Design A



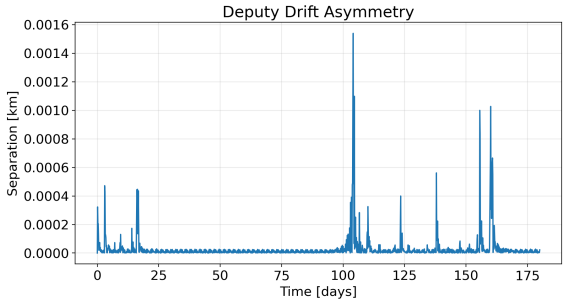
(a) Hill-frame trajectory of Design A with impulsive station-keeping maneuvers applied.



(b) Chief–deputy separation distance for Design A with impulsive corrections.



(c) Solar power input history for Design A with station-keeping maneuvers.



(d) Deputy asymmetry evolution for Design A under impulsive maintenance.

Figure 7: Impulsive formation maintenance results for Design A over six months.

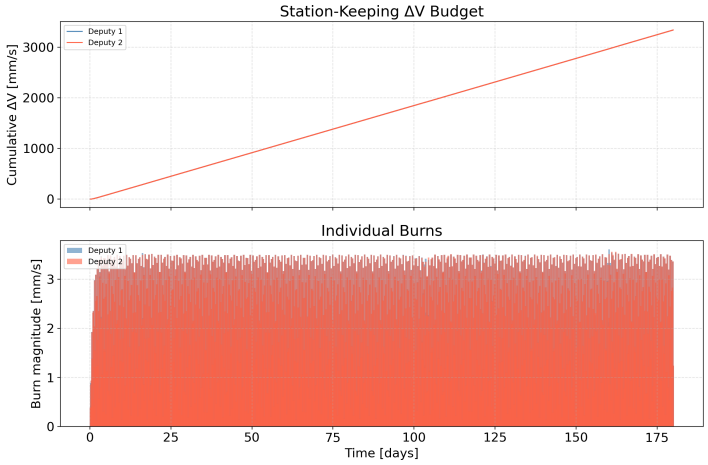
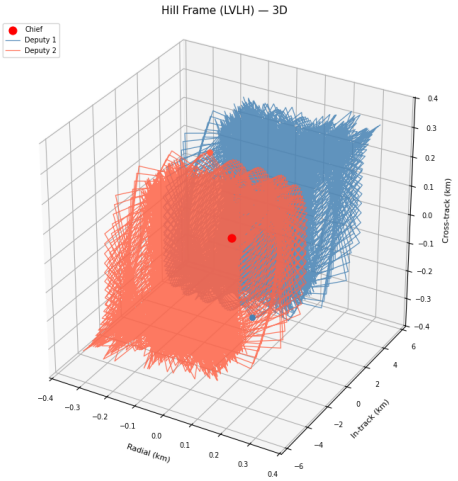
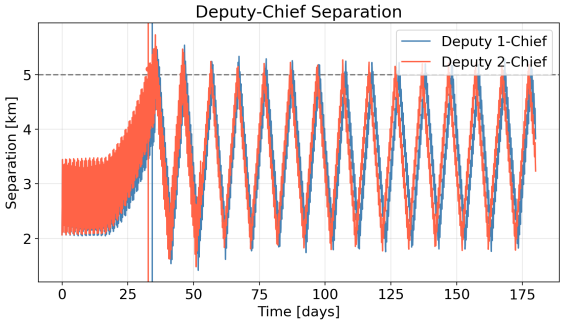


Figure 8: Delta V budget and burn times for design A

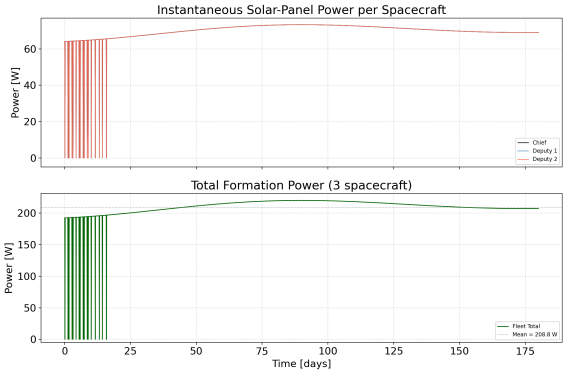
Design B



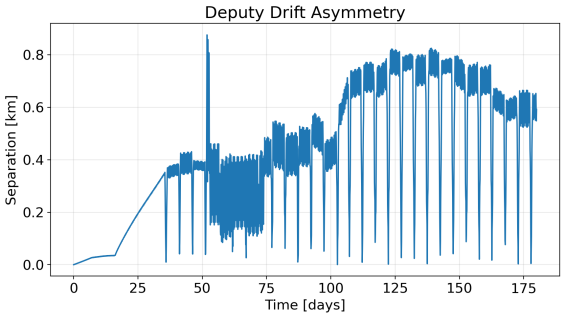
(a) Hill-frame trajectory of Design B with impulsive station-keeping maneuvers applied.



(b) Chief–deputy separation distance for Design B with impulsive corrections.



(c) Solar power input history for Design B with station-keeping maneuvers.



(d) Deputy asymmetry evolution for Design B under impulsive maintenance.

Figure 9: Impulsive formation maintenance results for Design B over six months.

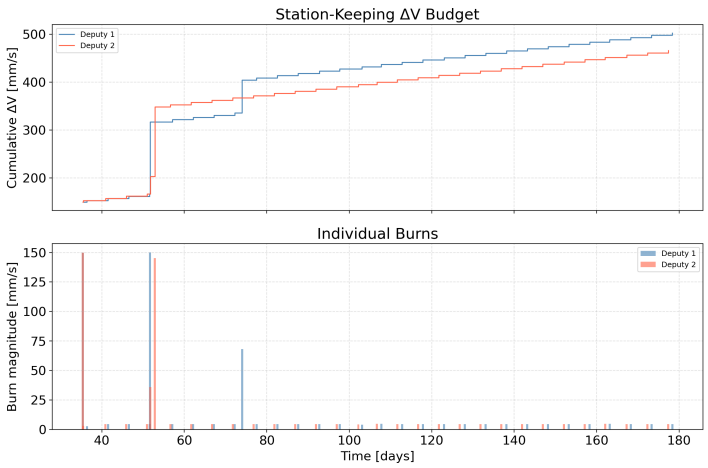


Figure 10: Delta V budget and burn times for design B

The impulsive maneuver analysis demonstrated that active station-keeping was capable of substantially reducing the secular drift introduced by the full force perturbation model. Figures 7 and 9 show the corrected deputy trajectories for both Design A and Design B after impulsive maneuvers were introduced into the propagation. In both configurations, the maneuvers successfully preserved the overall relative geometry of the formations and prevented the deputies from diverging significantly from the chief over the six-month simulation interval. Compared to the uncontrolled full-force trajectories, the maneuver-corrected formations remained considerably more bounded in the Hill frame and exhibited improved symmetry between the deputy spacecraft.

4 Discussion

Idealized Force Model - AP

Figures 3 and 4 show the evolution of inter-satellite distances for two formation designs under J_2 perturbations over a six-month period. In both cases, a clear secular increase in separation is observed, indicating that the formations are not strictly J_2-invariant.

Importantly, the rate of separation growth appears to be similar across both designs. While the absolute distances differ due to the initial formation size, the slopes of the distance curves are comparable, suggesting that the deputies experience nearly identical secular drift rates. However, what would likely be different across these two designs is the point where the deputies leave the bounding box set for each respective design.

Figure 4 provides additional insight into the relative motion geometry. In the Earth-centered frame, the deputies trace out dense, quasi-periodic trajectories around the chief. In the Hill (RTN) frame, the motion appears as tilted ellipses, with the radial–in-track projection forming closed curves and the in-track–cross-track projection showing

phase offsets between the deputies.

Despite the gradual drift observed in the simulations, the relative trajectories remained highly structured and symmetric throughout the propagation period. The deputies experienced nearly identical secular behavior, indicating that the chosen relative orbital element parameterization successfully minimized differential perturbation effects between spacecraft. This symmetry is operationally beneficial because it implies that both deputies would require similar station-keeping effort over the mission lifetime, simplifying maneuver planning and reducing the likelihood of uneven propellant depletion across the formation.

The results highlight that without enforcing strict J_2-invariance conditions, even well-structured formations will experience gradual separation growth, regardless of their initial scale.

Full Force Model - AP, AA

The full force model demonstrated the limitations of relying solely on J_2-invariant orbit design for long-duration formation flight in low Earth orbit. While the idealized simulations showed that the relative trajectories could remain bounded under only J_2 perturbations, introducing additional perturbations such as solar radiation pressure, atmospheric drag, higher-order gravity harmonics, and third-body effects from the Sun and Moon produced noticeable secular drift in both formation configurations.

The additional perturbations caused gradual distortion of the intended relative geometries, particularly in the in-track direction. Over the six-month propagation period, the deputy trajectories no longer remained perfectly closed in the Hill frame and instead exhibited slowly diverging quasi-periodic behavior. This confirmed that although J_2 invariance can suppress the dominant secular effects from Earth oblateness, it cannot fully cancel the combined influence of all environmental perturbations acting on spacecraft in SSO.

The degradation was significantly more pronounced for Design A than for Design B. Design A utilized a much smaller bounding box, meaning that even small perturbation-induced deviations became large relative to the formation size. As a result, the helix geometry distorted more rapidly and required substantially tighter control tolerances to preserve operational usefulness. In contrast, the larger along-track separations in the string-of-pearls configuration allowed Design B to tolerate greater absolute drift before violating mission constraints.

Another important result was that the perturbations affected the orbital elements differently. Variations in eccentricity and inclination introduced slow rotations and distortions in the deputy trajectories, while perturbations to the mean anomaly produced

growing along-track offsets over time. The cumulative effects of these perturbations motivated the need for active station-keeping strategies, which are explored further in the impulsive Stationkeeping analysis.

Impulsive Stationkeeping - AA

The impulsive maneuver analysis demonstrated that active station-keeping is necessary to preserve the desired formation geometries once higher-order perturbations are included in the model. Although the passive orbit design minimized secular drift from J_2, the additional perturbations accumulated over time and eventually caused the formations to violate their separation constraints without corrective action.

The maneuver strategies required for the two formation designs were fundamentally different due to the differences in formation geometry and allowable tolerances. Design A required many small corrective maneuvers because the tight helix formation could tolerate very little drift before the deputies deviated significantly from the intended relative geometry. Even relatively minor perturbations produced measurable distortions in the formation, requiring frequent station-keeping burns to maintain stability. While the magnitude of each burn was small, the high maneuver frequency increased operational complexity and cumulative control effort.

In contrast, Design B required fewer but more strategically timed maneuvers. Because the deputies were separated primarily in the along-track direction and operated within a larger allowable bounding box, the formation could tolerate significantly greater drift before correction became necessary. This reduced the required maneuver frequency and improved the long-term feasibility of the configuration for missions with limited onboard propellant.

The maneuver results also provided insight into propulsion system feasibility. The frequent, low-magnitude corrections required by Design A are more compatible with precision propulsion systems such as cold gas propulsion or Hall-effect thrusters. Meanwhile, Design B's lower maneuver frequency and larger allowable drift make propulsion systems such as hot gas propulsion or electrospray thruster arrays more practical options.

An additional observation from the maneuver analysis was that Design A may be better modeled in the future as a continuous low-thrust control problem rather than a purely impulsive one. Because the required corrections occur so frequently, continuous low-thrust actuation may provide a more efficient and realistic control strategy for maintaining extremely tight formations over long mission durations.

5 Orbital Stability and Analysis

Our orbital stability analysis showed that under only J2 perturbations, the formation designs were able to minimize secular drift effectively, allowing the deputies' relative orbits to remain closed and stable over time. However, when the model was expanded to include additional perturbations such as solar radiation pressure, atmospheric drag, and third-body effects from the Sun and Moon, the limitations of relying solely on J2 invariance became apparent. These added perturbations introduced significant secular drift that could no longer be naturally canceled by the orbit design alone. This effect was especially pronounced in Design A, where the smaller bounding box caused perturbations to have a much larger effect relative to the formation size, leading to faster geometric distortion of the formation. To counteract this drift, impulsive maneuvers were implemented to preserve the original formation geometry for as long as possible. The analysis showed that the two formation designs required fundamentally different station-keeping approaches. Design B was able to maintain stability with relatively few, well-spaced impulses, while Design A required many more frequent but smaller corrective maneuvers. This suggests that tighter formations are inherently more sensitive to perturbations, since even small deviations become significant enough to require immediate correction in order to maintain the intended geometry.

6 Design Feasibility

Our feasibility analysis showed that the two formation designs required different propulsion and control approaches based on their station-keeping demands. Design A required a much more active control scheme because its tighter formation geometry made it highly sensitive to orbital perturbations and drift. As a result, propulsion systems capable of frequent and precise corrections, such as cold gas propulsion or Hall effect thrusters, were identified as viable options for maintaining the formation. In contrast, Design B benefited from greater in-track separation between satellites, which naturally reduced the amount of station-keeping required and allowed the formation to remain stable with fewer corrective maneuvers. Because of these lower control demands, propulsion systems such as hot gas propulsion or arrays of electrospray thrusters were considered feasible solutions for maintaining the formation efficiently over time.

7 Summary and Future Work - AA

Our project evaluated the viability of a Chief–2 Deputy satellite formation system using two different formation designs. We first modeled the system under only J2

perturbations and later expanded the analysis to include additional natural orbital perturbations. The results showed that while secular drift caused by J2 effects could be minimized through careful orbit design, it was not possible to fully eliminate drift from all perturbations simultaneously. As a result, active station-keeping through impulsive maneuvers became necessary to maintain the desired formations over time. We then designed and analyzed the maneuver strategies required to preserve each formation configuration. One notable result was that the helix formation performed better at close proximity, while the string-of-pearls configuration became more effective for larger bounding boxes. Future work could focus on improving the station-keeping strategy for Design A, which currently requires frequent impulsive corrections at minute-scale intervals. Modeling this instead as a continuous low-thrust control problem could potentially achieve similar results more efficiently. Additionally, because orbital data centers would likely need operational lifetimes much longer than the six-month period considered in this study, future analyses should examine long-term orbital stability, fuel requirements, and maneuver optimization over extended mission durations.

8 Code Availability Statement and AI Disclosure

The code used to generate the plots and data referenced in this report can be found at https://github.com/GeostationaryBeaver/16.346-Astrodynamics/tree/main/term_project.

Additionally, the code used for this project was written with assistance from OpenAI's ChatGPT, Anthropic's Claude, and Google's Gemini. We have reviewed and edited the code as necessary for reasonableness and accuracy of the results produced. None of the analysis was written by AI.

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